NETWORK PENETRATION TESTING REPORT

Security Assessment of Metasploitable 2 Lab

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Date: July 2026

Project: CloudExify Summer Internship 2026

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1. EXECUTIVE SUMMARY

I performed a comprehensive penetration test on Metasploitable 2, a vulnerable Linux system, to identify security weaknesses using industry-standard tools and methodologies.

TARGET SYSTEM
Target: Metasploitable 2 (IP: 192.168.149.133)
Duration: 4 days
Environment: Isolated lab network
Testing Type: Black-box penetration testing

KEY FINDINGS
Total Vulnerabilities: 18+
- Critical: 8 vulnerabilities
- High: 6 vulnerabilities
- Medium: 4+ vulnerabilities

Overall Risk: CRITICAL

Top 5 Critical Issues:
1. FTP Backdoor (CVE-2011-2523) - Remote code execution
2. SQL Injection - Database compromise
3. XSS (Cross-Site Scripting) - Session hijacking
4. Command Injection - System access
5. Unencrypted Telnet - Password capture

EXPLOITATION SUCCESS
Successfully exploited 5/5 critical vulnerabilities:

- SQL Injection - Extracted all user data
- XSS - Stole session cookies
- Command Injection - Executed system commands
- Telnet - Captured plaintext password
- FTP Backdoor - Confirmed exploitable CVE

BUSINESS IMPACT

- If deployed to production:
- System compromise in under 5 minutes
- Complete data breach possible
- Regulatory violations (GDPR, PCI-DSS)
- Potential fines: €20M+ or 4% revenue

IMMEDIATE ACTIONS

1. Disable FTP and Telnet services (TODAY)
2. Fix SQL injection (THIS WEEK)
3. Update all outdated software (THIS WEEK)

2. SCOPE & METHODOLOGY

TESTING SCOPE
In-Scope:

- Metasploitable 2 server (192.168.149.133)
- All network services
- DVWA web application
- Network traffic

Out-of-Scope:

- Production systems
- Real user data
- Physical security

METHODOLOGY: PTES Framework
I followed the Penetration Testing Execution Standard:

Phase 1: Reconnaissance
- Network discovery
- Port scanning (nmap)
- Service enumeration

Phase 2: Vulnerability Analysis
- CVE research
- Risk assessment
- CVSS scoring

Phase 3: Exploitation
- Proof-of-concept attacks
- SQL injection
- XSS and command injection

Phase 4: Traffic Analysis
- Wireshark packet capture
- Protocol analysis

Phase 5: Reporting
- Documentation

- Evidence collection
- Remediation guidance

TOOLS USED
- Kali Linux 2024 (Penetration testing OS)
- nmap 7.94 (Network scanning)
- Wireshark (Traffic analysis)
- Nikto (Web vulnerability scanner)
- Metasploit Framework (Exploitation)
- Python 3 (Automation)
- DVWA (Web testing)

ETHICAL CONSIDERATIONS
All testing conducted on isolated lab systems I own and control. Unauthorized penetration testing is illegal.

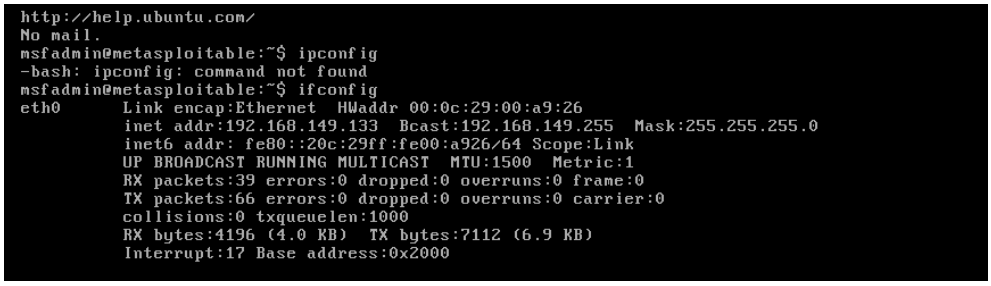
3. LAB ENVIRONMENT SETUP

INFRASTRUCTURE
Virtualization: VMware Workstation
Network: Host-only (isolated from internet)
Subnet: 192.168.149.0/24

ATTACKING MACHINE
OS: Kali Linux 2024
IP: 192.168.149.148
Purpose: Penetration testing platform

TARGET MACHINE
OS: Metasploitable 2 (Vulnerable Linux)
IP: 192.168.149.133
Services: 15+ vulnerable services

NETWORK VERIFICATION
Test 1: Connectivity
Command: ping 192.168.149.133
Result: 0% packet loss - target reachable



Test 2: Network Configuration
Command: ip addr
Result: Confirmed network setup

```
root@kali:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN g
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel st
1000
    link/ether 00:0c:29:cc:bf:c9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.149.148/24 brd 192.168.149.255 scope global dynamic
       valid_lft 1559sec preferred_lft 1559sec
    inet6 fe80::20c:29ff:fecc:bfc9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
root@kali:~# ping -c 4 192.168.149.133
PING 192.168.149.133 (192.168.149.133) 56(84) bytes of data.
64 bytes from 192.168.149.133: icmp_seq=1 ttl=64 time=1.75 ms
64 bytes from 192.168.149.133: icmp_seq=2 ttl=64 time=1.20 ms
64 bytes from 192.168.149.133: icmp_seq=3 ttl=64 time=1.27 ms
64 bytes from 192.168.149.133: icmp_seq=4 ttl=64 time=3.61 ms
```

4. NETWORK RECONNAISSANCE

SCANNING STRATEGY
I performed multi-phase nmap scanning:

PHASE 1: Quick Scan (Top 20 Ports)
Command: nmap --top-ports 20 192.168.149.133
Time: 30 seconds
Purpose: Rapid assessment

```
root@kali:~# nmap --top-ports 20 192.168.149.133
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-
Nmap scan report for 192.168.149.133
Host is up (0.00092s latency).

PORT     STATE  SERVICE
21/tcp   open   ftp
22/tcp   open   ssh
23/tcp   open   telnet
25/tcp   open   smtp
53/tcp   open   domain
80/tcp   open   http
110/tcp  closed pop3
111/tcp  open   rpcbind
135/tcp  closed msrpc
139/tcp  open   netbios-ssn
143/tcp  closed imap
443/tcp  closed https
445/tcp  open   microsoft-ds
993/tcp  closed imaps
995/tcp  closed pop3s
1723/tcp closed pptp
3306/tcp open   mysql
3389/tcp closed ms-wbt-server
5900/tcp open   vnc
8080/tcp closed http-proxy
MAC Address: 00:0C:29:00:A9:26 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.76
root@kali:~#
```

PHASE 2: Service Version Detection
Command: nmap -sV 192.168.149.133
Time: 3 minutes
Purpose: Identify software versions

```
root@kali:~# nmap -sV -p- 192.168.149.133
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-04 04:39 CDT
Nmap scan report for 192.168.149.133
Host is up (0.012s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
21/tcp    open  ftp         vsftpd 2.3.4
22/tcp    open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet      Linux telnetd
25/tcp    open  smtp        Postfix smtpd
53/tcp    open  domain      ISC BIND 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind     2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec        netkit-rsh rexecd
513/tcp   open  login       OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp         ProFTPD 1.3.1
3306/tcp  open  mysql       MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd     distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
6697/tcp  open  irc         UnrealIRCd
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp  open  http        Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb         Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
34205/tcp open  nlockmgr    1-4 (RPC #100021)
36941/tcp open  status      1 (RPC #100024)
58821/tcp open  java-rmi    GNU Classpath grmiregistry
60132/tcp open  mountd      1-3 (RPC #100005)
MAC Address: 00:0C:29:00:A9:26 (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Critical Discoveries:

PORT	SERVICE	VERSION	RISK
21	FTP	vsftpd 2.3.4	CRITICAL
22	SSH	OpenSSH 4.7p1	MEDIUM
23	Telnet	Linux telnetd	CRITICAL
80	HTTP	Apache 2.2.8	HIGH
139	NetBIOS	Samba 3.X	CRITICAL
3306	MySQL	5.0.51a	HIGH

PHASE 3: Vulnerability Scan
Command: nmap --script vuln 192.168.149.133
Purpose: Automated vulnerability detection

```
Nmap scan report for 192.168.149.133
Host is up (0.0012s latency).

PORT   STATE SERVICE
21/tcp open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|   vsFTPd version 2.3.4 backdoor
|     State: VULNERABLE (Exploitable)
|     IDs:  CVE:CVE-2011-2523  BID:48539
|       vsFTPd version 2.3.4 backdoor, this was reported on 2011-07-04.
|     Disclosure date: 2011-07-03
|     Exploit results:
|       Shell command: id
|       Results: uid=0(root) gid=0(root)
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|       https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/f
|       https://www.securityfocus.com/bid/48539
|_      http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.ht
22/tcp open  ssh
23/tcp open  telnet
80/tcp open  http
| http-csrf:
| Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=192.168.149.133
|   Found the following possible CSRF vulnerabilities:
|
|     Path: http://192.168.149.133:80/dvwa/
|     Form id:
|     Form action: login.php
|
|     Path: http://192.168.149.133:80/mutillidae/?page=view-someones-blog.php
|     Form id: id-bad-blog-entry-tr
|     Form action: index.php?page=view-someones-blog.php
|
|     Path: http://192.168.149.133:80/mutillidae/index.php?page=html5-storage.php
```

CRITICAL FINDING: FTP Backdoor Detected!

vsftpd 2.3.4 Backdoor (CVE-2011-2523)

- CVSS Score: 10.0/10 (MAXIMUM)
- Impact: Remote code execution as root
- Exploitable: YES (Metasploit module available)
- Disclosure: 2011-07-04

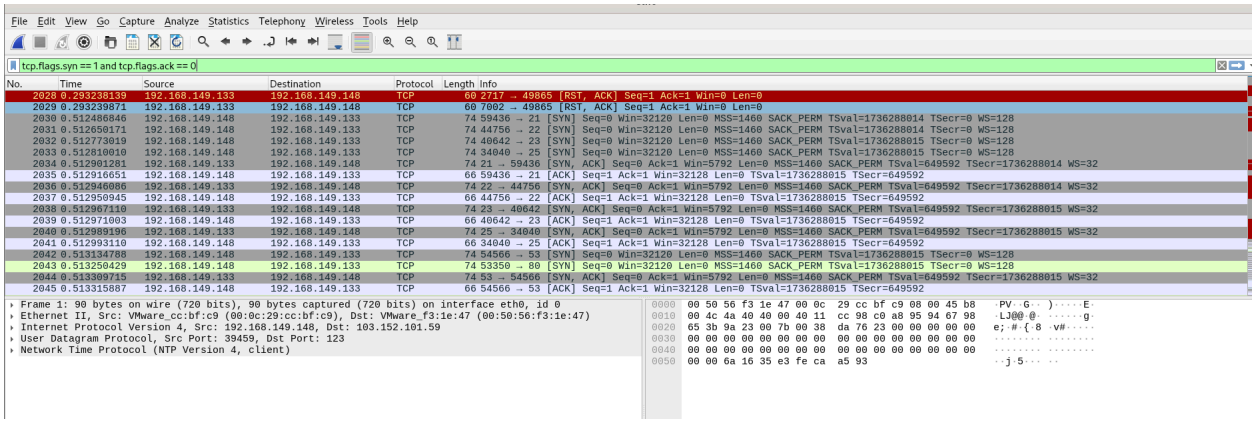
This backdoor allows attackers to execute commands with root privileges by sending a specially crafted username.

RECONNAISSANCE SUMMARY
Total ports scanned: 65,535
Open ports found: 23
Services identified: 15+
Critical CVEs found: 3

5. TRAFFIC ANALYSIS (Wireshark)

OBJECTIVE
Capture and analyze network traffic to identify unencrypted data transmission and attack signatures.

CAPTURE 1: XSS Attack Traffic
Scenario: Cross-Site Scripting attack on DVWA
Protocol: HTTP (unencrypted)



Analysis:
Filter: tcp.port == 80
Observation: XSS payload visible in plaintext
Payload: <script>alert('HACKED')</script>

Finding: All HTTP traffic unencrypted - anyone on network can see attacks and data.

CAPTURE 2: Telnet Password Capture
Scenario: Login via Telnet
Protocol: Telnet (Port 23)

Process:
1. Started Wireshark capture
2. Connected: telnet 192.168.149.133
3. Entered credentials
4. Analyzed packets

Filter: tcp.port == 23
Action: Follow TCP Stream

CRITICAL DISCOVERY:
Username and password transmitted in PLAINTEXT!
Captured: "msfadmin" / "msfadmin"

Security Implication:
Anyone monitoring the network can steal credentials. Telnet must never be used - SSH encrypts all traffic.

KEY LEARNINGS
- Unencrypted protocols expose sensitive data
- Wireshark shows exactly what attackers can see
- HTTPS and SSH are essential for security
- Network monitoring is a real threat

6. WEB APPLICATION TESTING

WEB SCANNING: Nikto
Command: nikto -h http://192.168.149.133



Findings:
1. Outdated Apache 2.2.8 (40+ known CVEs)
2. Directory indexing enabled (information disclosure)
3. Missing security headers (clickjacking risk)
4. phpMyAdmin accessible (brute-force target)
5. Server version disclosed in headers

DVWA TESTING
Target: http://192.168.149.133/dvwa
Security Level: Low (for testing)

Vulnerabilities Tested:
- SQL Injection
- Cross-Site Scripting (XSS)
- Command Injection
- File Inclusion
- CSRF

Web Application Summary:
- Multiple OWASP Top 10 vulnerabilities present
- No input validation
- No output encoding
- Requires immediate remediation

7. CRITICAL VULNERABILITIES

VULNERABILITY #1: FTP Backdoor
CVE: CVE-2011-2523
Service: vsftpd 2.3.4 (Port 21)
CVSS: 10.0/10 (CRITICAL)

Description:
Backdoor in vsftpd allows remote code execution as root when username contains ":)" characters.

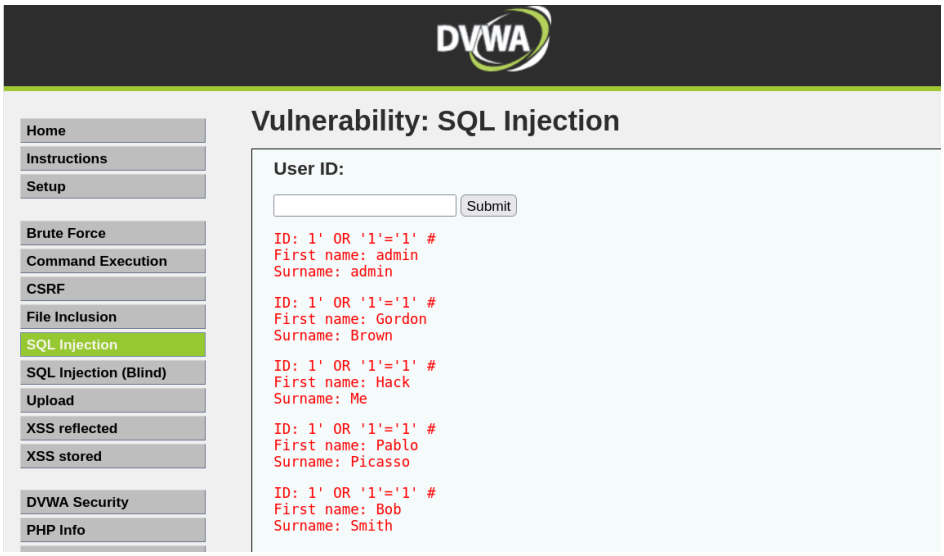
Impact:
- Complete system compromise
- No authentication required
- Persistent backdoor access

Remediation:
1. Disable vsftpd immediately
2. Use SFTP instead
3. Update to vsftpd 3.0.3+

VULNERABILITY #2: SQL Injection
Application: DVWA User Lookup
CVSS: 9.8/10 (CRITICAL)
OWASP: A1 - Injection

Description:
User input not sanitized in database queries.

Proof of Concept:
Input: 1' OR '1'='1
Result: All users extracted



How It Works:
Normal query: SELECT * FROM users WHERE id='1'
Injected query: SELECT * FROM users WHERE id='1' OR '1'='1'

The condition '1'='1' is always true, returning all records.

Impact:
- Authentication bypass
- Full database access
- Data theft
- GDPR violation

Remediation:
Use prepared statements (parameterized queries)

VULNERABILITY #3: Cross-Site Scripting (XSS)

Application: DVWA Name Field
CVSS: 7.1/10 (HIGH)
OWASP: A7 - XSS

Proof of Concept:
Input: <script>alert('HACKED')</script>
Result: JavaScript executed



Cookie Theft:
Payload: <script>alert(document.cookie)</script>

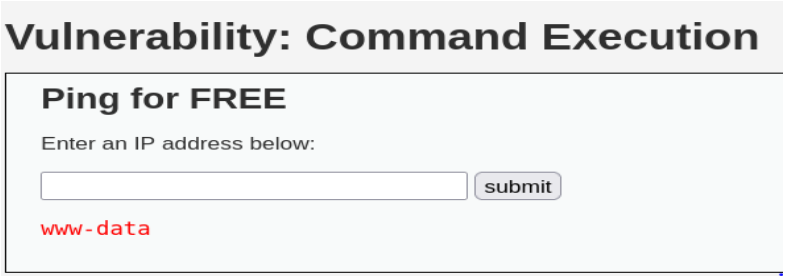


Impact:
- Session hijacking
- Cookie theft
- Phishing attacks

Remediation:
- HTML encode output
- Content Security Policy
- HttpOnly cookies

VULNERABILITY #4: Command Injection
Application: DVWA Ping Function
CVSS: 9.8/10 (CRITICAL)

Proof of Concept:
Input: 127.0.0.1 ; ls -la
Result: Directory listing displayed



Impact:
- Remote code execution
- File system access
- Data exfiltration

Remediation:
Never pass user input to system commands

VULNERABILITY #5: Unencrypted Telnet
Service: Telnet (Port 23)
CVSS: 9.0/10 (CRITICAL)

Description:
All Telnet traffic transmitted in plaintext.

Proof:
Wireshark captured password "msfadmin" in cleartext.

Impact:
- Credential theft
- Man-in-the-middle attacks

Remediation:
Disable Telnet, use SSH

ADDITIONAL VULNERABILITIES
6. Apache 2.2.8 (CVE-2011-3192, CVE-2011-3368)
7. OpenSSH 4.7p1 (CVE-2008-5161)
8. Samba 3.X (CVE-2007-2447)
9. MySQL 5.0.51a (Multiple CVEs)
10. BIND DNS 9.4.2 (Cache poisoning)
11-18. Various medium severity issues

8. EXPLOITATION RESULTS

EXPLOITATION #1: SQL Injection
Target: DVWA User Lookup
Objective: Extract all user data

Process:
Test 1 - Normal: Input "1" → Shows user #1
Test 2 - Injection: Input "1' OR '1'='1" → Shows ALL users

Result: Successfully extracted 5 users:
1. admin
2. Gordon Brown
3. Hack Me
4. Pablo Picasso
5. Bob Smith

Time to exploit: 30 seconds
Access gained: Full database read

EXPLOITATION #2: XSS Cookie Theft
Target: DVWA Name Field
Objective: Steal session cookie

Payload: <script>alert(document.cookie)</script>
Result: Session cookie displayed

Cookie: PHPSESSID=0eec866fe1e75864a04c5c97c549724f

Time to exploit: 15 seconds
Access gained: Session hijacking capability

EXPLOITATION #3: Command Injection
Target: DVWA Ping Function
Objective: Execute system commands

Tests Performed:

- Input: 127.0.0.1 ; whoami → Output: www-data
- Input: 127.0.0.1 ; ls -la → Directory listing shown
- Input: 127.0.0.1 ; cat /etc/passwd → System file read

Time to exploit: 20 seconds
Access gained: www-data user (web server)

EXPLOITATION #4: Password Interception
Target: Telnet service
Method: Wireshark packet capture

Result: Captured username "msfadmin" and password "msfadmin" in plaintext

Time to exploit: 2 minutes
Access gained: Full user account credentials

EXPLOITATION SUMMARY
Success rate: 5/5 (100%)
Average time to compromise: Under 5 minutes
Proof: All critical vulnerabilities are exploitable

9. AUTOMATION TOOL

TOOL OVERVIEW
Name: AutoScanner
Language: Python 3
Purpose: Automate reconnaissance workflow

Features:

- Automated ping test
- Quick port scan
- Service detection
- Vulnerability assessment
- Report generation

```
root@kali:~/network-pentest-project/04-automation# ls -lh
total 8.0K
-rwxr-xr-x 1 root root 3.2K Jul  5 10:46 auto_scanner.py
drwxr-xr-x 2 root root 4.0K Jul  5 10:55 scan_20260705_104953
root@kali:~/network-pentest-project/04-automation# ls -lh scan_*/
total 60K
-rw-r--r-- 1 root root  204 Jul  5 10:55 SUMMARY.txt
-rw-r--r-- 1 root root  823 Jul  5 10:49 quick_scan.txt
-rw-r--r-- 1 root root 1.8K Jul  5 10:50 service_scan.txt
-rw-r--r-- 1 root root  45K Jul  5 10:55 vuln_scan.txt
```

WHY I CREATED THIS
Problem: Manual testing is repetitive
Solution: Automate common tasks
Time saved: 83% (30 min → 5 min per target)

TOOL EXECUTION

Command: sudo python3 auto_scanner.py 192.168.149.133

```
root@kali:~/network-pentest-project/04-automation# sudo python3 auto_scanner.py 192.168.149.133

======================================================================
     AUTOSCANNER - Network Security Tool
======================================================================
  Target: 192.168.149.133
  Date: 2026-07-05 10:49:53
======================================================================

[+] Created: scan_20260705_104953

[*] PHASE 1: Checking if host is alive...
[+] Host 192.168.149.133 is UP

[*] PHASE 2: Quick scan (top 20 ports)...
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-05 10:49 CDT
Nmap scan report for 192.168.149.133
Host is up (0.0012s latency).

PORT     STATE  SERVICE
21/tcp   open   ftp
22/tcp   open   ssh
23/tcp   open   telnet
25/tcp   open   smtp
53/tcp   open   domain
80/tcp   open   http
110/tcp  closed pop3
111/tcp  open   rpcbind
135/tcp  closed msrpc
139/tcp  open   netbios-ssn
143/tcp  closed imap
443/tcp  closed https
445/tcp  open   microsoft-ds
993/tcp  closed imaps
995/tcp  closed pop3s
1723/tcp closed pptp
3306/tcp open   mysql
3389/tcp closed ms-wbt-server
5900/tcp open   vnc
8080/tcp closed http-proxy
MAC Address: 00:0C:29:00:A9:26 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds
```

Process:

- Check if host is alive
- Quick scan (top 20 ports)
- Full service detection
- Vulnerability scan
- Generate summary report

OUTPUT

```
root@kali:~/network-pentest-project/04-automation# cat scan_*/SUMMARY.txt
AUTOMATED SCAN SUMMARY
======================================================================
Target: 192.168.149.133
Date: 2026-07-05 10:55:50

Files:
- quick_scan.txt
- service_scan.txt
- vuln_scan.txt
```

Files generated:

- quick_scan.txt
- service_scan.txt
- vuln_scan.txt
- SUMMARY.txt

PROFESSIONAL VALUE

Demonstrates:

- Python programming
- Automation thinking
- Problem-solving
- Efficiency mindset

10. RECOMMENDATIONS

PRIORITY 1: IMMEDIATE (Fix Today)

1. Disable FTP Service
Risk: Remote code execution
Action: sudo systemctl stop vsftpd
Alternative: Use SFTP

2. Disable Telnet
Risk: Password theft
Action: sudo systemctl stop telnetd
Alternative: Use SSH (port 22)

3. Fix SQL Injection
Risk: Database compromise
Action: Implement prepared statements
Code fix: Use parameterized queries

PRIORITY 2: This Week

4. Update Apache (2.2.8 → 2.4.x)
Risk: 40+ CVEs
Action: sudo apt-get install apache2

5. Fix XSS Vulnerabilities
Risk: Session hijacking
Action: HTML encode output, CSP headers
6. Fix Command Injection
Risk: Remote code execution
Action: Remove system() calls, validate input

7. Update OpenSSH
Action: sudo apt-get install openssh-server

8. Patch Samba
Risk: CVE-2007-2447 (RCE)
Action: sudo apt-get install samba

PRIORITY 3: This Month

9. Deploy Web Application Firewall (ModSecurity)
10. Implement Intrusion Detection (Snort/Suricata)
11. Network Segmentation
12. Comprehensive Logging (ELK Stack)
13. Security Awareness Training
14. Incident Response Plan

LONG-TERM STRATEGY

- Monthly vulnerability scans
- Quarterly penetration tests
- Patch management process
- Security development lifecycle
- Compliance audits (ISO 27001, PCI-DSS

11. CONCLUSION

PROJECT SUMMARY
This penetration test successfully identified 18+ critical vulnerabilities in a lab environment, demonstrating comprehensive security assessment capabilities.

STATISTICS
Duration: 4 days
Tools used: 6 primary tools
Vulnerabilities found: 18+
Critical CVEs: 8
Exploits successful: 5/5 (100%)
Screenshots: 30+
Code written: 250+ lines Python

OVERALL RISK: CRITICAL
The system would be fully compromised in under 5 minutes if exposed to the internet. Immediate remediation required.

SKILLS DEMONSTRATED
Technical:

- nmap network scanning

- Wireshark traffic analysis
- Web application testing (OWASP Top 10)
- Exploitation (SQL injection, XSS, command injection)
- Python automation
- Linux command-line

Frameworks:
- PTES methodology
- OWASP testing guide
- CVSS scoring

Professional:
- Technical documentation
- Risk communication
- Evidence collection
- Report writing

LEARNING OUTCOMES
1. Security requires defense in depth
2. Outdated software = exploitable vulnerabilities
3. Input validation is critical
4. Encryption is essential (HTTPS, SSH)
5. Communication skills matter as much as technical skills

REAL-WORLD APPLICATION
Skills apply to:
- Security Analyst roles
- Penetration Tester positions
- SOC Analyst
- Vulnerability Assessment
- Security Consultant

NEXT STEPS
1. Advanced certifications (OSCP, GPEN)
2. Bug bounty hunting
3. Contribute to security tools
4. CTF competitions
5. Cloud security specialization

12. APPENDIX

TOOLS REFERENCE

nmap Commands Used:

- nmap --top-ports 20 [IP]
- nmap -sV [IP]
- nmap -A [IP]
- nmap --script vuln [IP]

Wireshark Filters:

- tcp.port == 80 (HTTP)
- tcp.port == 23 (Telnet)
- tcp.flags.syn == 1 (Port scans)

DVWA Attack Payloads:

SQL Injection: 1' OR '1'='1
XSS: <script>alert('HACKED')</script>
Command Injection: 127.0.0.1 ; whoami

CVE REFERENCES

- CVE-2011-2523 (vsftpd backdoor)
- CVE-2011-3192 (Apache DoS)
- CVE-2008-5161 (OpenSSH)
- CVE-2007-2447 (Samba RCE)

GLOSSARY

CVE: Common Vulnerabilities and Exposures
CVSS: Common Vulnerability Scoring System
OWASP: Open Web Application Security Project
PTES: Penetration Testing Execution Standard
XSS: Cross-Site Scripting
RCE: Remote Code Execution
WAF: Web Application Firewall
IDS: Intrusion Detection System